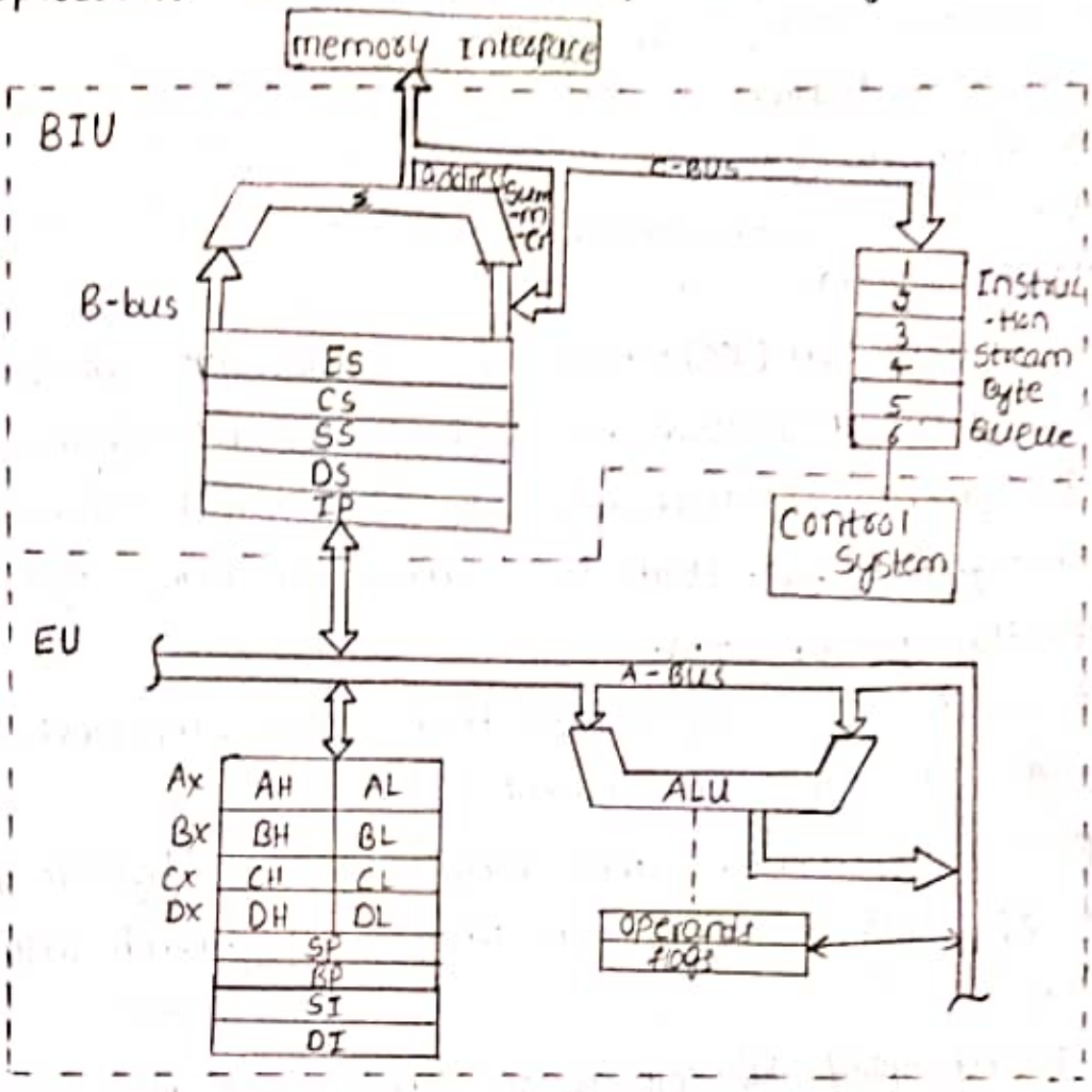
MPMC

MODULE - I

1 Explain the internal hardware architecture of 8086 microprocessor with the help of block diagram?

A



The internal architecture of intel 8086 is divided into 2 units:

→ The Bus interface unit (BIU)

→ The execution unit (EU)

the Bus interface unit:

It provides the interface of 8086 to external memory and I/O devices via the system bus.

It generates the 20-bit physical address for memory access.

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It fetches instructions from the memory
It transfer data to and from the memory and I/O.
maintains the 6-byte pre-fetch instruction queue (suppo-rts pipelining).

BIU contains mainly 4 segment registers.
→ Instruction pointer
→ pre-fetch queue
→ Address generation circuit

Segment Registers:

Code segmentation [CS]: Holds base address for code segment
Data segment [DS]: Holds base address for data segment
Stack segment [SS]: Holds base address for stack segment
Extra segment [ES]: Holds base address for extra segment

Instruction pointer (IP):

It is a 16-bit register. It holds offset - the next inst-ruction in the code segment

IP is incremented after every instruction byte is fetch-ed. IP gets a new value whenever a branch instruction occurs.

CS is multiplied by 10H to give the 20-bit physical addre-ss of the code segment

The address of the next instruction is calculated by using the formula [CS x 10H + IP]

Pre-fetch queue:

→ It is a 6-byte queue [FIFO]. because the maximum size of instruction that can have in 8086 is 6 bytes.
→ fetching the next instruction while executing the current instruction (pipelining)

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→ It is connected with the control unit which is responsible for decoding op-code & operands & telling EU what to do with the help of timing & CS [control signals].

→ It is responsible for pipelining

Address generation circuit :

→ The BIU has a physical address generation circuit

→ It generates 20 bit physical address using segment & offset addresses using formula.

[physical address = segment address x 10H + offset address]

Execution unit [EU] :-

EU tells the BIU from where to fetch instructions or data, decodes instructions & executes instructions

It contains → control circuitary
→ Instruction decodes
→ ALU
→ Register organization
↳ flag register
↳ general purpose register
↳ pointer & index register

→ the control circuitary in the EU directs the internal operations.

→ A decodes can translate the instructions fetched from the memory into a series of actions, which the EU performs.

→ ALU is 16-bit it can add, subtract, AND, OR, XOR, increment, decrements, complement & shift binary number

→ In registers organization general purpose registers

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are used to held data results temperorily.

→ flag register can control the status of the processor, pointers & index registers are associated with segment registers to produce 20-bit address.

Q What is microprocessor and list the architectural fea-tures of 8086 microprocessor

A Microprocessor is a digital integrated circuit that act as central processing unit of a computer, handling arthimatic, logic, control and I/O operations.

(or)

the microprocessor is an electronic chip that functions as the central processing unit (CPU) of a computer.

→ It is essentially the brain of a computer executi-ng instructions from programs to perform various task.

Features of 8086:

→ It is a 16 bit microprocessor having 20 address lines and 16 data lines and provides upto 1MB storage

→ It has an instruction queue, which is capable of storing six instruction bytes from the memory resulting in faster processing.

→ It was the 1st 16 bit processor having 16 bit ALU, 16 bit registers, internal data bus and 16 bit external data bus resulting in faster processing

→ It is available in 3 versions based on frequency of operation.

8086 → 5 MHz

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8086 - 2 → 8MHz

8086 - 1 → 10MHz

→ It uses two stages of pipelining
1. fetching stage
2. Executing stage
which improves performance

→ fetch stage can prefetch upto 6 bytes of instructions & stores them in the queue

→ Execute stage executes these instructions

→ It consists of 29000 transistors.

→ It was designed by intel 1976

→ It operates in two modes
* Maximum mode
* Minimum mode

→ It operates at 5V

→ It has powerful instruction set

→ It has two units
* Bus Interface unit
* Execution unit

3 Explain the register organization of the 8086 microprocessor?

A Register organization :-

The 8086 has a powerful set of registers. It includes general purpose registers, segment registers, pointers and index registers and flag registers.

General purpose registers :-

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AX	AH	AL
BX	BH	BL
CX	CH	CL
DX	DH	DL

The 8086 has 4 16-bit general purpose registers labeled AX, BX, CX and DX

Each 16-bit general purpose register can be split into '2' 8-bit registers.

The letters L & H refers to the lower & higher bytes of particular register & X refers total 16-bit register [L & H are of each 8-bit]

AX (Accumulator):-

AX is used as 16-bit accumulator, The lower 8-bit of AX represented as AL and higher 8-bits of AX represe-nted as AH

It is used to store the result of some arithmetic operations.

BX (Base register):-

BX holds the offset address of a location in the me-mory. It is also used to refer to the data in the memory

It is used as offset storage for generating physical addresses in case of certain addressing modes.

CX (Count register):-

CX is default count register used to hold the count value while executing the repeated string instructions and the loop instructions.

CL is used to hold the count value while executing

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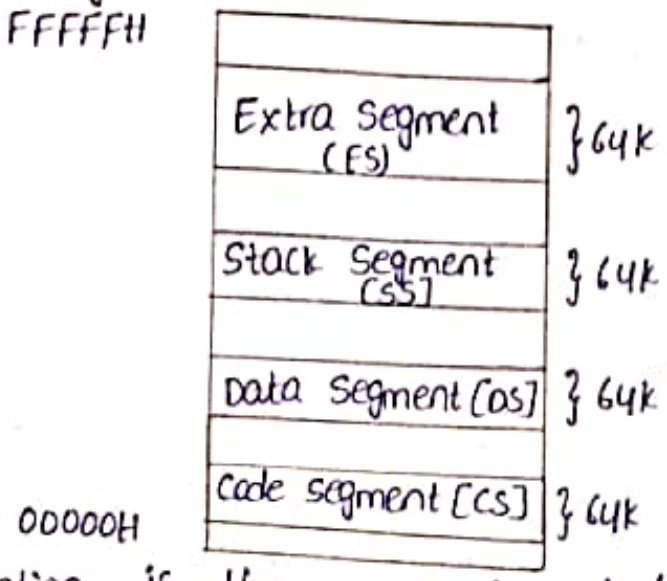
the Shift / rotate instruction.

Dx (Data register) :-

Dx register which may be used as an operand / destination incase of few instructions.

This register can be used in multiplication, input / output addressing etc

Segment registers :-



Segmentation is the process in which the main memory of the computer is logically divided into different segments & each segment has its own base address.

It is used to increase the speed of execution and processor is able to fetch and execute the data from the memory easily & fast

Code segmentation register [CS] :-

It is used for addressing memory location in the code segment of the memory, where the executable program is stored.

Data segmentation register [DS] :-

It points to the data segment of the memory where the data is stored

Stack segment register (SS) :-

It is used for addressing stack segment of the memory.

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The stack segment is that segment of memory which is used to store stack data.

Rules of segmentation:-

The starting address of a segment should be such that it can be evenly divided by 16.

Minimum size of a segment can be 16 bytes and the maximum can be 64KB.

Pointers and Index register

SP
BP
SI
DI
IP

The pointers will always store some address (or) memory location in 8086 microprocessor, they usually stores the offset through which the actual address is calculated

Instruction pointer (IP):-

The instruction pointer usually stores the address of the next instruction that is to be executed

Apart from this, it also acts as an offset for CS register.

Base pointer (BP):-

The base pointer stores the address of the memory. Also, it acts as an offset for stack segment (SS)

Stack pointer (SP):-

The stack pointer points at the current top value of the stack. Like the BP it also acts as an offset to the stack segment (SS). SP with SS register can generate 20-bit physical address.

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Source Index [SI]:-

It is used to store the offset of source data in data segment

Destination Index [DI]:-

It is used to store the offset of destination in data (or) extra segment

4. Draw the pin configuration of 8086 microprocessor and explain the following pins

i) $AD_0 - AD_{15}$ ii) MN/MX' iii) NMI iv) ALE v) M/IO' vi) DT/R

A 8086 can operate in any one of the following two modes - minimum & maximum modes.

It is packaged in a 40 pin DIP package. The pin configuration shown in below figure

8086 signals can be categorised in 3 groups.

1) signals having common functions in min & max mode

2) signals which have special functions for min mode

3) signals which have special functions for max mode

i) $AD_0 - AD_{15}$:-

These pins act as address-data bus. whenever ALE (Address Latch Enable) is set (logic 1), these pins carry address, and when ALE is at '0' these pins carry data. If we want to demultiplex address, data bus seperately by using latches along with ALE.

→ A_{16}/S_3 to A_{19}/S_6 :- These pins are multiplexed to provide address signals ($A_{16} - A_{19}$) & status bits ($S_3 - S_6$). when ALE = 1

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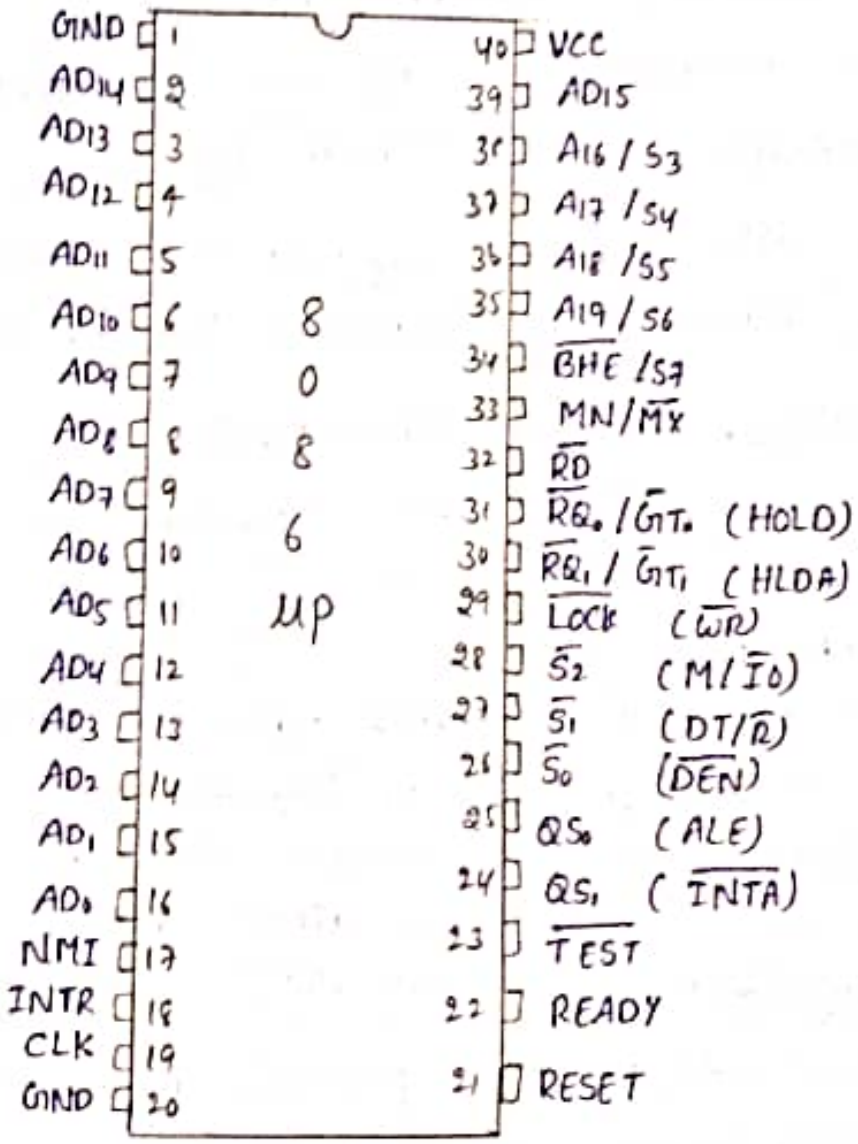
these pins carry address, ALE=0 these pins carry data.
In status bus S_0 & S_4 indicate the segment accessed by the 8086 during the current bus cycle

S_4	S_3	accessed
0	0	Extra segment
0	1	Stack segment
1	0	Code / No segment
1	1	Data segment

Pin Diagram



→ $\overline{BHE}$ / S_7 :- The bus high enable is used in the 8086 to enable the higher data bus (D_8 - D_{15}) during read/ write operation. S_7 is always at logic 1.

ii) MN/$\overline{MX}$:-

This pin is used to select either minimum or maximum mode configuration. when this pin is connected to power supply (+5V) it operates at minimum mode, when it is connected to ground it operates in maximum mode

iii) NMI :-

The Non-mastable interrupt is a hardware interrupt. when it occurs, the type 02H interrupts occurred

iv) ALE :-

When the address latch enable is at high indicates address/data bus & address/status bus contain address which can be either memory / IO address.

v) M/$\overline{IO}$:-

this pin indicates whether the 8086 is performing memory read/write operation (M/$\overline{IO}$ = 1) or I/O read/write operation (M/$\overline{IO}$ = 0).

vi) DT/$\overline{R}$:-

The data transmit or receive signal indicates that the 8086 data bus transmitting (DT/$\overline{R}$ = 1) or receiving (DT/$\overline{R}$ = 0) data.

This signal is used to control the data flow direction in external data bus.

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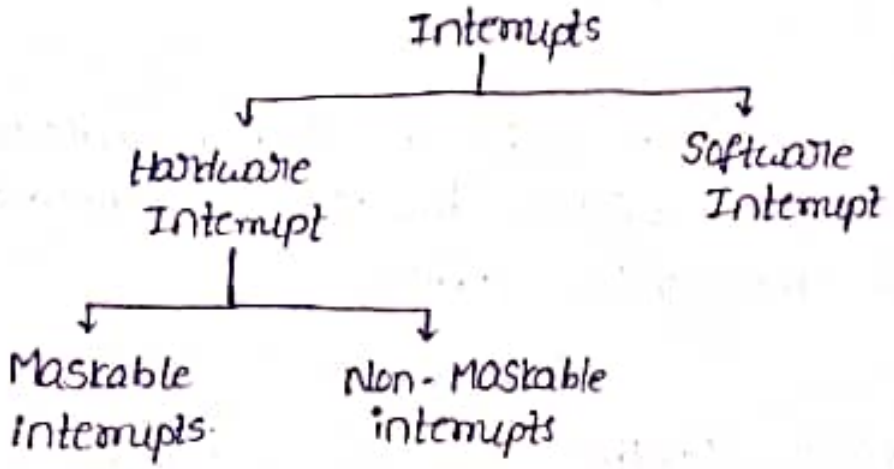
5 What is Interrupt and types? Explain the concept of interrupt structure of an 8086 microprocessor and write about interrupt vector table with diagram?

A Interrupt :-

Interrupt is the method of creating a temporary halt during program execution and allows peripheral devices to access the microprocessor

The microprocessor responds to that interrupt with an ISR (Interrupt Service routine). which is a short program to instruct the microprocessor on how to handle the interrupt

Types of Interrupts



Hardware interrupts :-

Hardware interrupts is caused by an pheripheral device by sending a signal through a specific pin to the microprocessor.

The 8086 has two hardware interrupts pins i.e, NMI and INTR.

NMI is a non-maskable interrupt having lower priority.

INTA called interrupt acknowledge.

NMI :-

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It is a single non-mastable interrupt pin having lower priority than the mastable interrupt request pin.

When this interrupt is activated then it:

→ Completes the current instruction that is in progress

→ pushes the CS (Code Segment) value and IP (Instruction Pointer) value of the return address on to the stack

→ IP is loaded from the contents of the word location 00008H

→ CS is loaded from the contents of the next word location 0000AH

→ Interrupt flag and trap flag are reset to 0.

INTR:

The INTR is a mastable interrupt. It is an interrupt activated by an I/O port. If the interrupt is enabled and NMI is disabled, then the microprocessor first completes the current execution and sends '0' on INTA pin twice.

Actions taken by microprocessor

→ 1st completes the current instruction

→ Activates INTA output and receives the interrupt type.

→ flag register value, CS value of the return address & IP value of the return address are pushed on to the stack

→ IP value is loaded from the contents of word location X × 4.

→ CS is loaded from the contents of the next word location.

→ Interrupt flag and trap flag is reset to 0.

Software Interrupts:

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The execution of the instruction INT n; where n is the interrupt type that can take any value between 00H to FFH. This is called Software interrupt

Types of Software interrupts:

Type 0:

whenever the results from a division overflow or an attempt is made to divide by 'zero'.

Type 1: Single step interrupt

when the 8 bit of the flag register is set to 1, an interrupt no. 1 is generated after the execution of every instruction.

Type 2:

The non-maskable interrupt occurs when logic 1 is placed on the NMI pin to the up. non-maskable it cannot be disabled

Type 3:

A special one-byte instruction (INT 3) that uses this vector to access its interrupt device procedure. often used to store a breakpoint in a program for debu-gging.

Type 4:

overflow is a special vector used with the INTO instruction. The INTO interrupts the program if an overflow condition exists.

Interrupt response (Interrupt structure):-

After execution of each instruction in a program the 8086 checks if any interrupt has been requested. If an interrupt has been requested, the 8086

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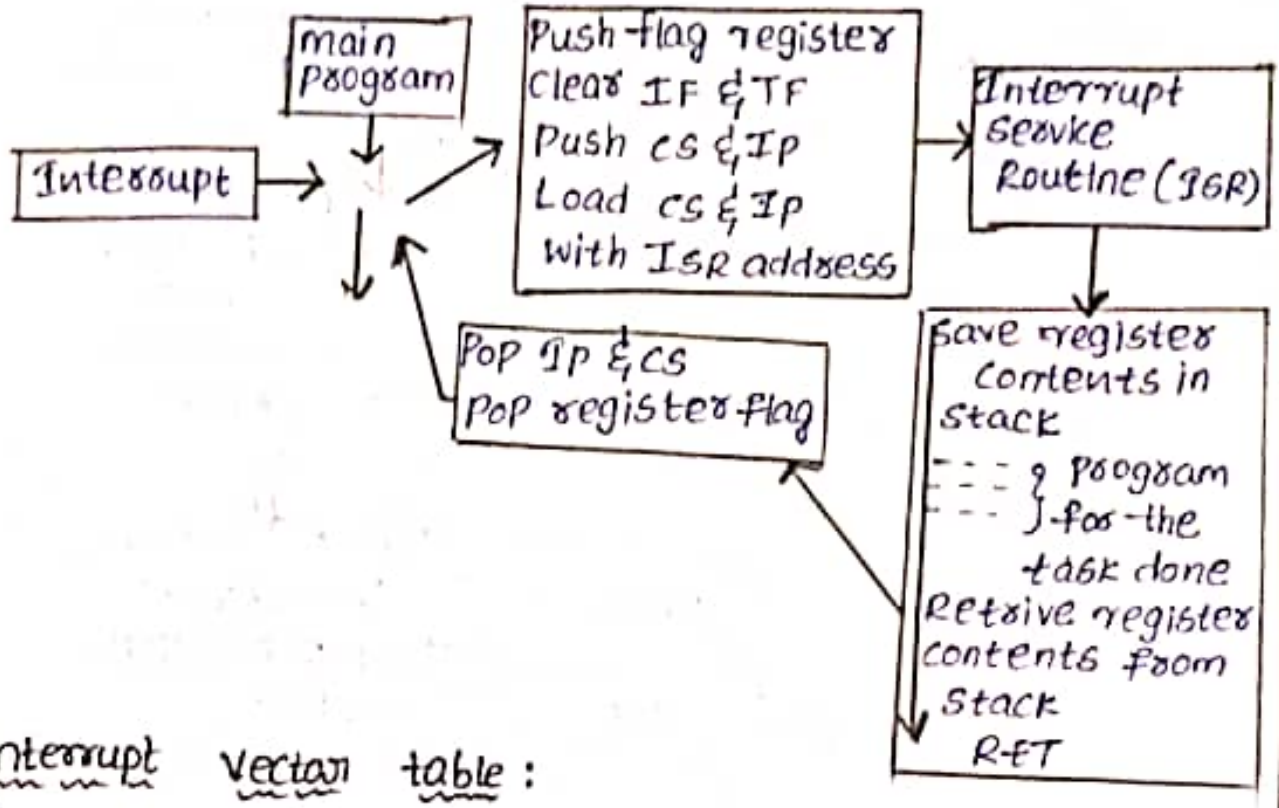
processor it by performing the following series of the steps.



Interrupt vector table :

The first 1k byte of memory of 8086 (00000 to 003FF) is set aside as a table for storing the starting address of interrupt service procedure (ISP).

In the table, IP value is put in as low word of the vector & CS is put to high vector.

The interrupt vector table as shown in below.

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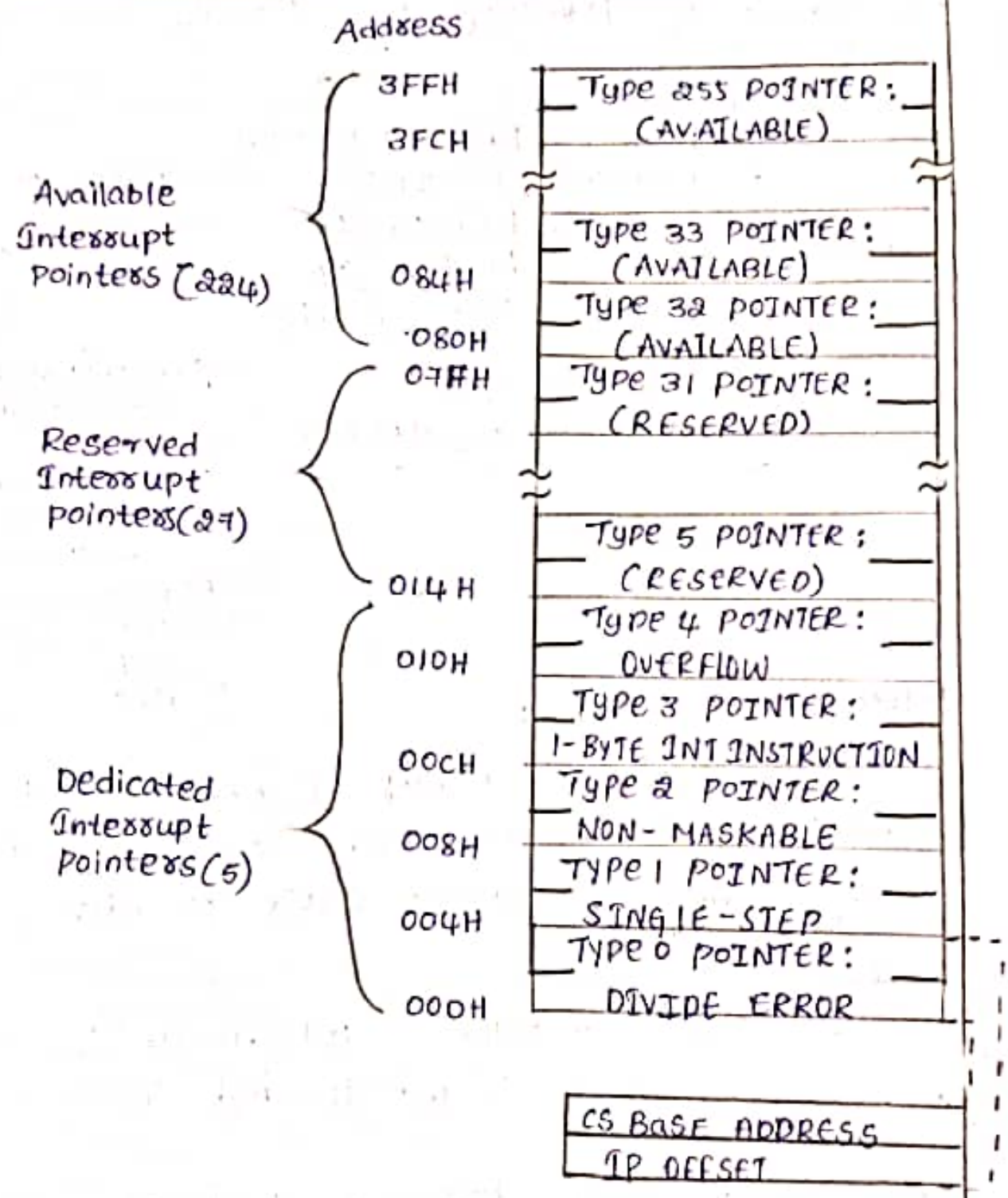

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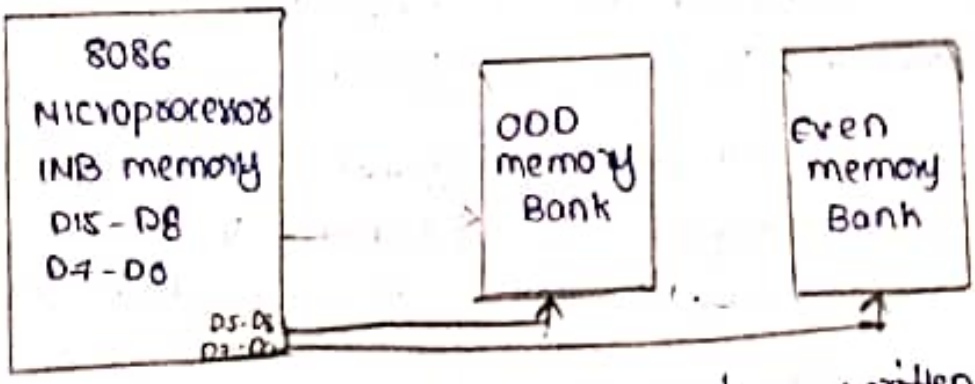
6) A) Explain with diagram about physical memory organization?

Ans:- Memory organization:-

There are 20 address lines in the 8086 Microprocessor. This gives us 220 different memory location. Hence the total size is 220 bytes (as each memory location is Byte addressable, i.e one byte of data can be stored at every single location). which is equal to 1MB.

8086 microprocessor can easily handle up to 16-bits of data at a time through it's 16- data lines. So, to organize the memory efficiency, the entire memory in 8086 is divided into two memory banks i.e; 1) odd bank
2) Even bank



The way in which data is used as written is decided by in which the value of BHE, and the last address bit, that is the A0 line, It is done in following way:

BHE (Bus high Enable)

BHE	A0	operation performed on memory
0	0	operation performed on memory 16-bit of data will be used written into the memory
0	1	8 bit of data will be read/written into odd memory
1	0	8 bit of data will be read/written into memory bank (Even)
1	1	No operation is performed.

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→ To read/write 8 bit of data, it would require only 1 CPU cycle. no matter the data is stored in any of the memory banks, but to read/write 16-bits of data, the BIU of the 8086 may require 1 or 2 memory cycles depending upon whether the lower byte of word is located at even/odd memory address.

→ If LB of word is stored at even memory bank & the UB is stored at odd memory bank then the CPU will require only 1 memory cycle.

→ If LB of would is located at an odd memory. then CPU needs 2 memory cycle. 1st cycle for accessing data of LSB through higher data Bus (D_{15} - D_8).

→ 2nd cycle is required for accessing the upper byte of the data through the lower bus (D_7 - D_0)

3) Describe the function of flag register with example?

Ans:- The 8086 flag register contents indicate the result of computing in the ALU. It also contain some flag bits to control the CPU operations.

A 16-bit flag register is used in 8085. It is divided into two parts : → condition code / status flags

→ Machine control flag.

The condition code flag register is the lower byte of the 16-bit flag register. The condition code flag register is identical to 8085 flag register, with an additional overflow flag.

The control flag register is the higher byte of the flag register. It contains three flag namely : 1) Direction flag (D)

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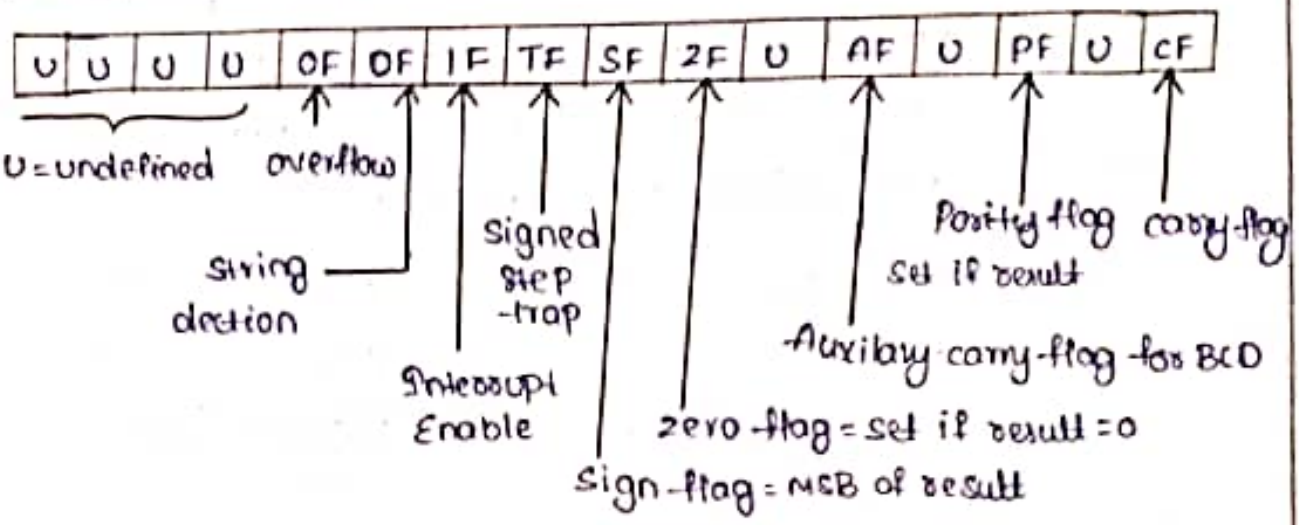
2) Interrupt flag (I)

3) Trap flag (T)

flag register configuration :-



SF - Sign flag : This flag is set, when the result of any computation is negative. For signed computation the SF equals the MSB of the result.

Ex:- 106 H - 0 (sign) 0 0 0 0 1 1 0

107 H - 0 0 0 0 0 1 1 1

(-)

sign 1 1 1 1 1 1 1 1 ⟹ SF = 1

ZF - Zero flg : This flag is set, if the result of the configuration (or) composition performed by the perious instruction is zero.

Ex:- 06 H - 0 0 0 0 0 1 1 0

06 H - 0 0 0 0 0 1 1 0

0 0 0 0 0 0 0 0 H $\boxed{ZF = 1}$

PF - Parity flag : This flag is set to 1, if the lower byte result contains even number of 1's.

Ex:- A 4 H - 1 0 1 0 0 1 0 0

9 5 H - 1 0 0 1 0 1 0 1

PF = 1 0 0 1 1 1 0 0 1 ⇒ [illegible]

no. of 1's = 4

CF - carry flag : This flag is set, when there is a carry out of MSB in case of addition / a borrow in case of subtraction.

Ex:- A 4 H - 1 0 1 0 0 1 0 0

9 5 H - 1 0 0 1 0 1 0 1

1 0 0 1 1 1 0 0 1 ⇒

carry flag

AF - Auxilary flag : This is set, if there is carry from the lowest nibble i.e; bit three during addition or borrow for the lowest nibble during subtraction

Ex:- 2 B H - 0 0 1 0 1 0 1 1

4 7 H - 0 1 0 0 0 1 1 1

1

0 1 1 1 0 0 1 0

AF = 1

OF - overflow flag : This flag is set, if the result of a signed operation is large to store in destination register.

Ex:- + 4 H - 0 1 0 0

+ 4 H - 0 1 0 0

1 0 0 0

↳ overflow because MSB represented sign

OF = 1

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TF - Trap flag : If this flag is set, the processor enters the single step execution mode.

The processor executes the current instruction and the control is transfered to the trap interrupt service mode.

IF - Interrupt flag : If this flag is set, the mask able interrupt are recongnized by the cpu, otherwise they are ignored.

D - Direction flag : If is used by staring manipulation instruction of this flag is set to '0'. the string is processed begining from lower lower address to high address i.e auto increment mode, other-wise auto content mode (i.e from high to low).

7) What are the functional block of 8086 internal architecture and explain each block?

Ans:- The 8086 microprocessor architecture is divided into two main functional blocks i.e; 1) the Bus interface unit (BIU)
2) The Execution unit (EU)

These units work in parallel to enhance processing speed and system throughput.

→ Bus Interface unit (BIU) :-

This unit is the 8086's interface to the outside world. It provides a full 16-bit bi-directional data bus and 20-bit address bus. The bus interface unit can perform all external bus operations like memory/I/O read/write operation.

Main Block in BIU :

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1) Instruction Queue (6 Byte):

→ Stores pre-fetched instructions (pipelining).

→ improves execution speed by overlapping fetch and execution.

2) Segment Registers (CS, DS, SS, ES):

→ used for memory segmentation (20-bit physical address generation).

- CS : code segment - for instruction
- DS : Data segment - for stack
- ES : Extra segment - for additional data.

3) Instruction pointer (IP):

→ Holds the offset address of the next instruction in the code segment.

4) Address Generation circuit:

→ combines segment and offset to form a 20-bit physical address.

physical Address = (segment x 10H) + offset

2) Execution unit (EU):

The EU responsible for executing instruction, performing arithmetic and logic operations, and managing the processor's internal operation.

Main Blocks in EU:

1) Arithmetic logic unit (ALU):

→ Performs all arthmetic (add, sub, mul, div) and logical (AND, OR, NOT, XOR) operation.

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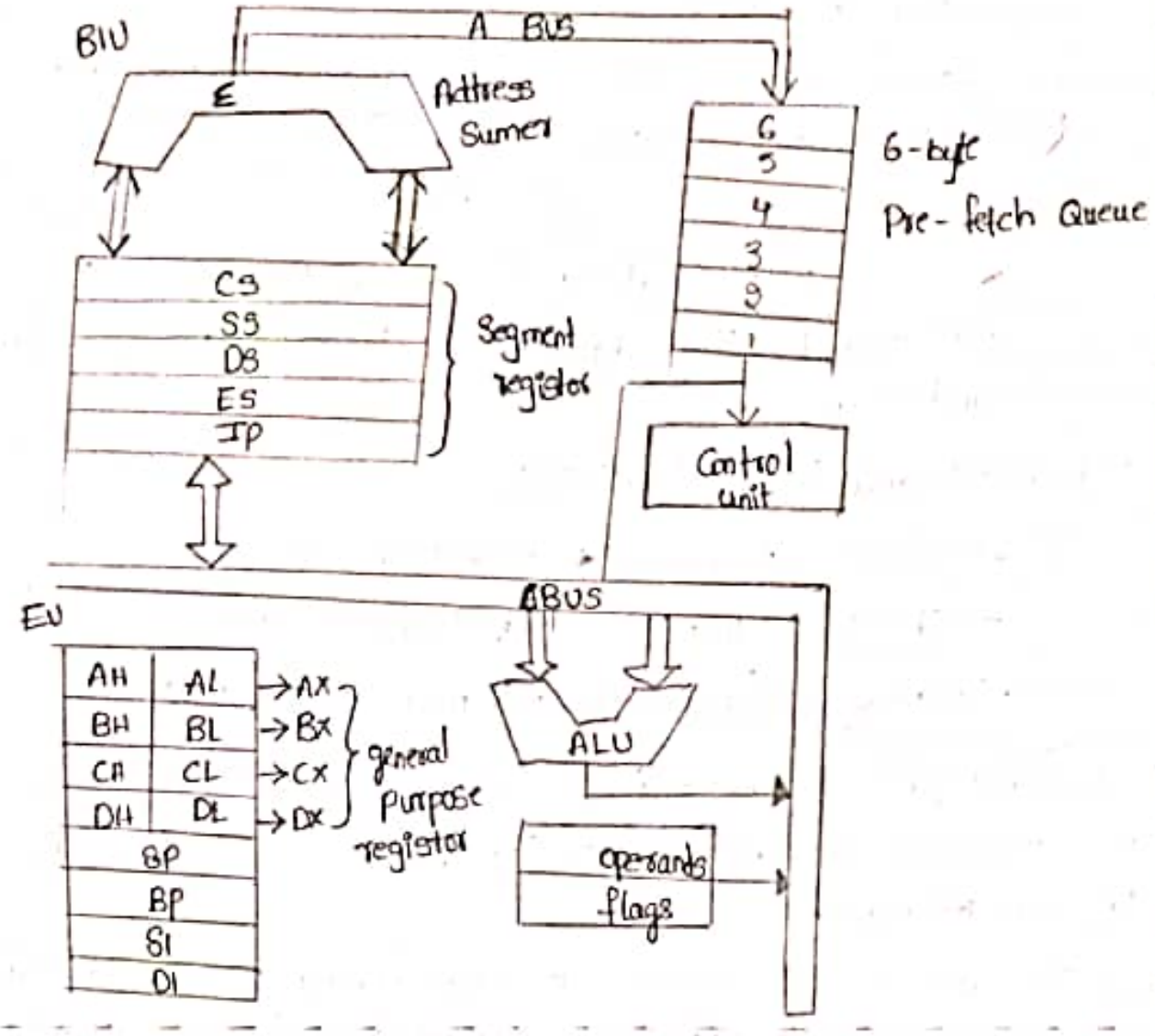

2) General purpose Registers (Ax, Bx, Cx, Dx):

→ used for temporary data storage during execution

- Ax : Accumulator
- Bx : Base Register
- Cx : Counter Register
- Dx : Data Registers

3) flag Register:

→ Indicates the Status of operations via flags:

- zero, sign, carry, overflow, parity, auxiliary carry, Direction, Interrupt Enable flag.

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4) Instruction Decoder:

→ Decodes instruction fetched by the BIU and prepares control signals for execution.

5) Control unit:

→ Directs internal operation of the EU.

→ controls the data flow between registers, ALU, and memory.

6) Pointers and Index Registers:

→ SP (Stack pointer, BP (Base pointer)

→ SI (Source Index), DI (Destination Index)

→ Mainly used for addressing memory and data movement.

8) What are the segment, pointer & index registers in 8086 microprocessor and how they mapped together to calculate Physical address?

Ans:- 1) In the 8086 microprocessor, the memory is organized using a segmented addressing scheme, where the 1MB memory space is divided into segments of maximum size 64KB.

2) To access a location, the processor uses two 16-bit values: a segment pointer and an offset. (effective address).

3) The segment pointer determines the starting address of the segment, while the offset indicates the distance from this start.

4) The four main segment registers (also called segment pointers): 1) code segment (CS)

~~2) Instruction seg~~

2) Data segment (DS)

3) Extra segment (ES)

4) Stack segment (SS)

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code segment (CS) which is used to holds the base address for instruction code.

→ Data segment (DS) is used for general data & stack segment (SS) is used for holds the base address for the stack, and Extra segment (ES) is used for string and extra data operations.

→ Along with these, 8086 has pointer and index registers for addressing inside a segment.

→ Base pointer (BP) and stack pointer (SP) are mainly used with the stack segment, while source index (SI) and Destination Index (DI) are commonly used in string and array operations.

→ Base registers (BX, BP) and Index registers (SI, DI) can also be combined in addressing modes to calculate the effective offset.

5) The physical address in 8086 is obtained by shifting the segment the register value left by 4 bits (multiplying by 16) and then adding the 16-bit offset from a pointer or index registers.

<u>For</u> Example:-

if DS = 1000H and SI = 2000H,

the segment base is 10000H (after left shift), and adding the offset 2000H gives the physical address 12000H.

This scheme allows overlapping segments and flexible addressing, enabling the processor to access a full 20-bit address space from 16-bit registers.

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9) Draw the timing diagrams of making maximum mode of 8086 microprocessor in read and write cycle?

Ans:-

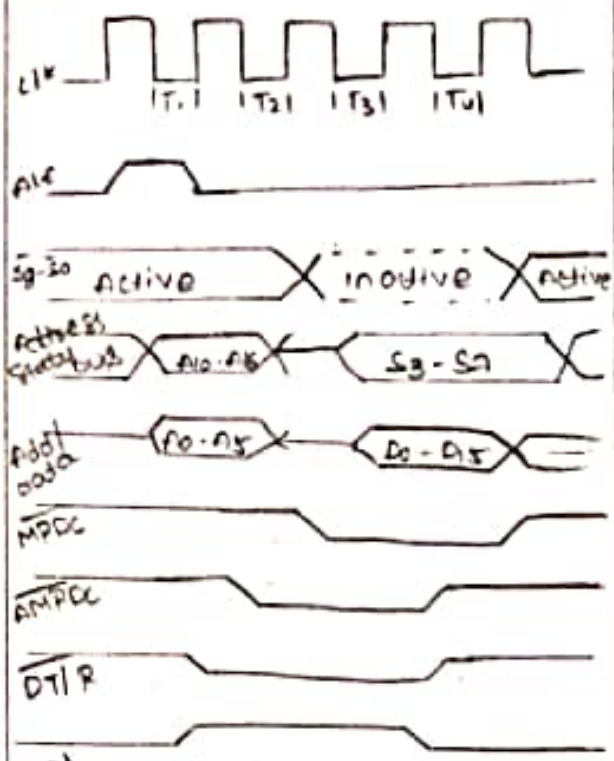

fig:- Read cycle

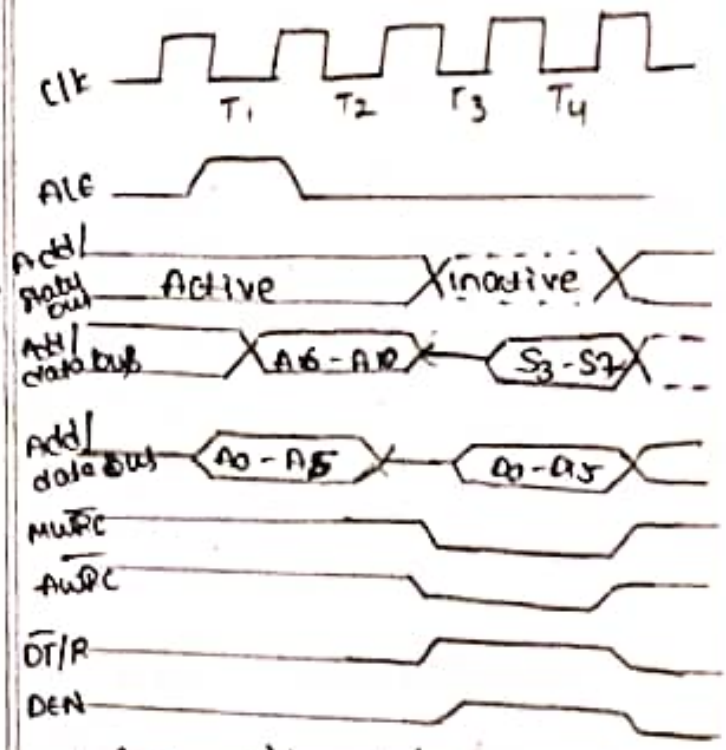

fig: write cycle.

In maximum mode, the 8086 works with the 8288 bus controller, which generates control signals from the status lines S_2, S_1, S_0. The bus controller issues command like MRDC, MWTC, IORC and IOWC for memory and I/O operation.

In read cycle, during T_1 the address is placed on A_{00} - A_{015} and latched using ALE. The 8288 activates DEN in T_2 and asserts MRDC/IORC to enable the addressed device to place data on the bus. In T_3 valid data is read, and in T_4 control signals are deactivated. wait states are inserted b/w T_3 and T_4 if Ready is low.

In the write cycle, T_1 is similar for address output. In T_2 the processor place data on the bus, and the 8288 issues advanced write signals (AMWTC/AIOWC). In T_3, MWTC/IOWC is active while data remains stable, and in T_4 all signals are deactived and the bus is freed. The reliable data transfer b/w CPU, memory.

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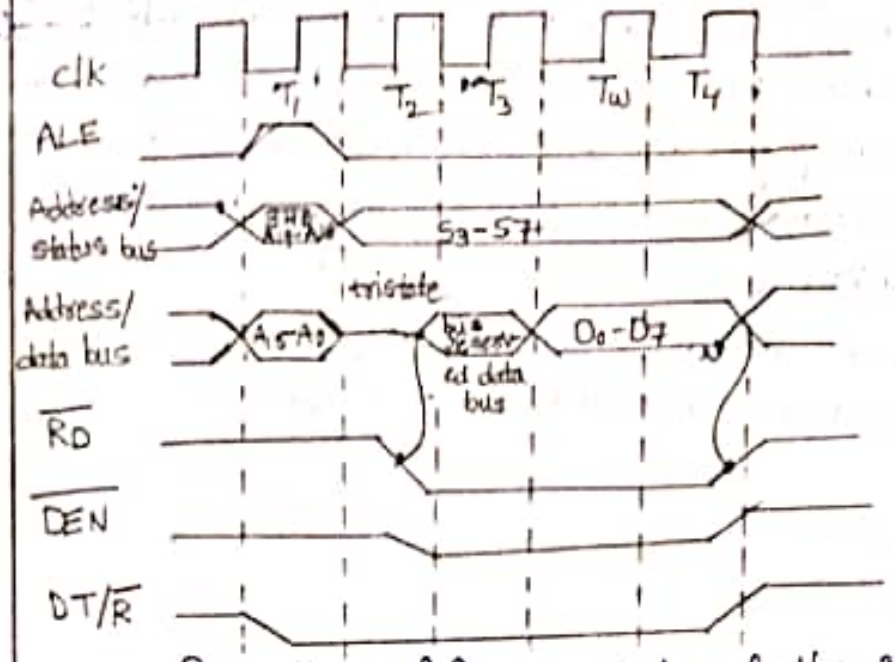
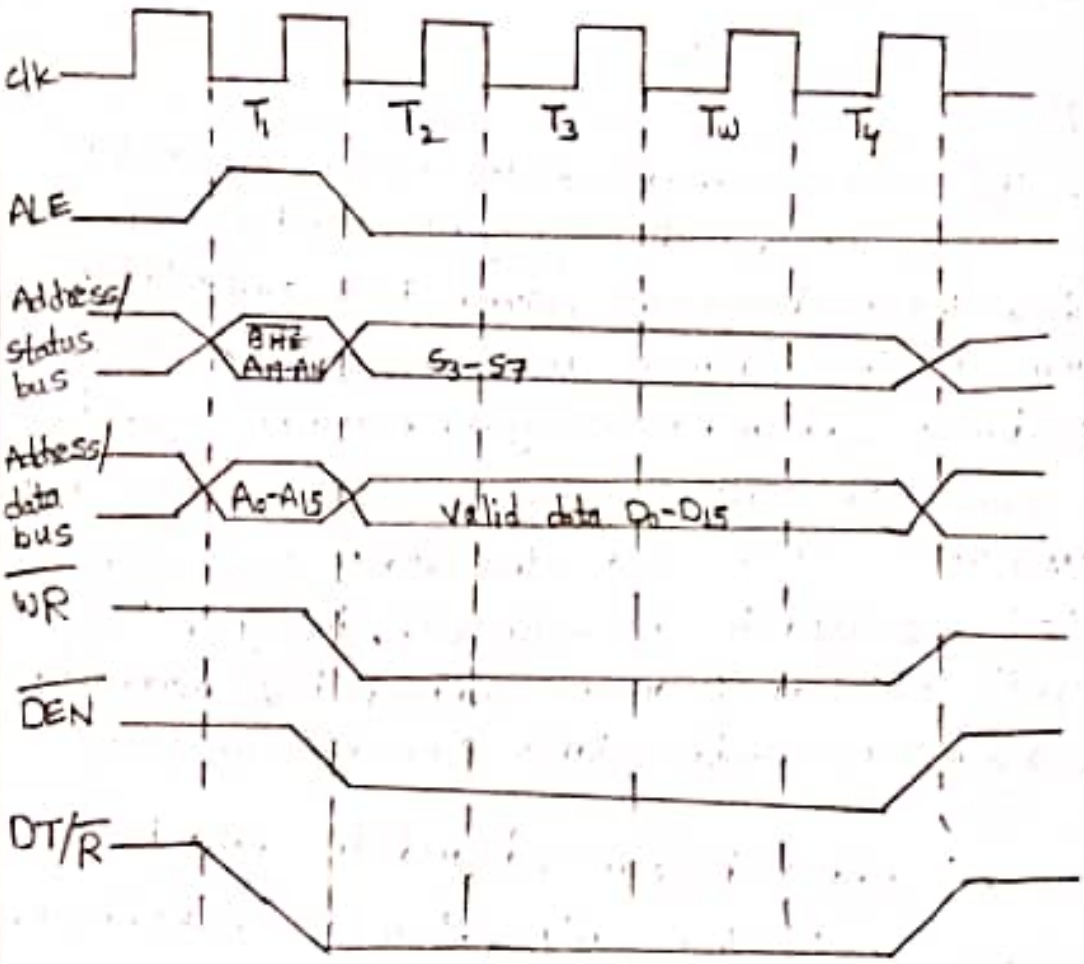
10) Draw the timing diagrams of minimum mode of 8086 microprocessor in read and write cycle ?.

Ans:



In the minimum mode of the 8086 microprocessor, all the control signals are directly generated by the processor itself without the need for an external bus controller. The timing diagram of the minimum mode shows the sequence of signals during read and write cycle. A machine cycle is divide into four clock states, namely T_1, T_2, T_3 and T_4, and in some cases, wait states (T_w) are inserted if slower memory or I/O devices are used. In the read cycle, during T_1 the processor outputs the address on the address bus along with the status signals ALE (Address latch enable) to latch the address.

In T_2, the address is removed and the data bus is floated for input, while the control signals RD (Read and DEN (Data enable) become active. During T_3, the addressed memory or I/O devices places the required data on the data bus. In T_4, the processor samples the data and RD goes inactive, completing the cycle.

Scanned with OKEN Scanner

In the write cycle, during T_1 the address and ALE are provided to latch the address, while in T_2 the processor place data on the data bus and activates WR (write). In T_3, the valid data is available on the bus and is written into the memory or I/O device. Finally, in T_4 the WR signal is deactived, ending the write cycle. Thus, the timing diagram clearly illustrates how the control, address, and data signals are synchronized with clock states to perform reliable read and write operations in minimum mode.